

2.2 Compound Statements



Figure 2.6 A person seeking their driver's license must pass two tests. A compound statement can be used to explain performance on both tests at once. (credit: modification of work "Drivers License -Teen driver" by State Farm/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Translate compound statements into symbolic form.
2. Translate compound statements in symbolic form with parentheses into words.
3. Apply the dominance of connectives.

Suppose your friend is trying to get a license to drive. In most places, they will need to pass some form of written test proving their knowledge of the laws and rules for driving safely. After passing the written test, your friend must also pass a road test to demonstrate that they can perform the physical task of driving safely within the rules of the law.

Consider the statement: "My friend passed the written test, but they did not pass the road test." This is an example of a **compound statement**, a statement formed by using a **connective** to join two independent clauses or logical statements. The statement, "My friend passed the written test," is an independent clause because it is a complete thought or idea that can stand on its own. The second independent clause in this compound statement is, "My friend did not pass the road test." The word "but" is the connective used to join these two statements together, forming a compound statement. So, did your friend acquire their driving license?

This section introduces common logical connectives and their symbols, and allows you to practice translating compound statements between words and symbols. It also explores the order of operations, or dominance of connectives, when a single compound statement uses multiple connectives.

Common Logical Connectives

Understanding the following logical connectives, along with their properties, symbols, and names, will be key to applying the topics presented in this chapter. The chapter will discuss each connective introduced here in more detail.

The joining of two logical statements with the word "and" or "but" forms a compound statement called a **conjunction**. In logic, for a conjunction to be true, all the independent logical statements that make it up must be true. The symbol for a conjunction is $\wedge$. Consider the compound statement, "Derek Jeter played professional baseball for the New York Yankees, and he was a shortstop." If p represents the statement, "Derrick Jeter played professional baseball for the New York Yankees," and if q represents the statement, "Derrick Jeter was a short stop," then the conjunction will be written symbolically as $p \wedge q$.

The joining of two logical statements with the word "or" forms a compound statement called a **disjunction**. Unless otherwise specified, a disjunction is an inclusive *or* statement, which means the compound statement formed by joining two independent clauses with the word *or* will be true if a least one of the clauses is true. Consider the compound statement, "The office manager ordered cake for for an employee's birthday or they ordered ice cream." This is a disjunction because it combines the independent clause, "The office manager ordered cake for an employee's birthday," with the independent clause, "The office manager ordered ice cream," using the connective, *or*. This disjunction is true if

the office manager ordered only cake, only ice cream, or they ordered both cake and ice cream. Inclusive *or* means you can have one, or the other, or both!

Joining two logical statements with the word *implies*, or using the phrase “if *first statement*, then *second statement*,” is called a **conditional** or *implication*. The clause associated with the “if” statement is also called the **hypothesis** or *antecedent*, while the clause following the “then” statement or the word *implies* is called the **conclusion** or *consequent*. The conditional statement is like a one-way contract or promise. The only time the conditional statement is false, is if the hypothesis is true and the conclusion is false. Consider the following conditional statement, “If Pedro does his homework, then he can play video games.” The hypothesis/antecedent is the statement following the word *if*, which is “Pedro does/did his homework.” The conclusion/consequent is the statement following the word *then*, which is “Pedro can play his video games.”

Joining two logical statements with the connective phrase “if and only if” is called a **biconditional**. The connective phrase “if and only if” is represented by the symbol, $\leftrightarrow$. In the biconditional statement, $p \leftrightarrow q$, p is called the hypothesis or antecedent and q is called the conclusion or consequent. For a biconditional statement to be true, the truth values of p and q must match. They must both be true, or both be false.

The table below lists the most common connectives used in logic, along with their symbolic forms, and the common names used to describe each connective.

Connective	Symbol	Name
and but	$\wedge$	conjunction
or	$\vee$	disjunction, inclusive or
not	$\sim$	negation
if ..., then implies	$\rightarrow$	conditional, implication
if and only if	$\leftrightarrow$	biconditional

Table 2.3



These connectives, along with their names, symbols, and properties, will be used throughout this chapter and should be memorized.

EXAMPLE 2.8

Associate Connectives with Symbols and Names

For each of the following connectives, write its name and associated symbol.

1. or
2. implies
3. but



Solution

1. A compound statement formed with the connective word *or* is called a disjunction, and it is represented by the $\vee$ symbol.
2. A compound statement formed with the connective word *implies* or phrase “if ..., then” is called a conditional statement or implication and is represented by the $\rightarrow$ symbol.
3. A compound statement formed with the connective words *but* or *and* is called a conjunction, and it is represented by the $\wedge$ symbol.

> YOUR TURN 2.8

For each connective write its name and associated symbol.

1. not
2. and
3. if and only if

Translating Compound Statements to Symbolic Form

To translate a compound statement into symbolic form, we take the following steps.

1. Identify and label all independent affirmative logical statements with a lower case letter, such as p , q , or r .
2. Identify and label any negative logical statements with a lowercase letter preceded by the negation symbol, such as $\sim p$, $\sim q$, or $\sim r$.
3. Replace the connective words with the symbols that represent them, such as $\wedge$, $\vee$, $\rightarrow$, or $\leftrightarrow$.

Consider the previous example of your friend trying to get their driver's license. Your friend passed the written test, but they did not pass the road test. Let p represent the statement, "My friend passed the written test." And, let $\sim q$ represent the statement, "My friend did not pass the road test." Because the connective *but* is logically equivalent to the word *and*, the symbol for *but* is the same as the symbol for *and*; replace *but* with the symbol $\wedge$. The compound statement is symbolically written as: $p \wedge \sim q$. My friend passed the written test, but they did not pass the road test.

 When translating English statements into symbolic form, do not assume that every connective "and" means you are dealing with a compound statement. The statement, "The stripes on the U.S. flag are red and white," is a simple statement. The word "white" doesn't express a complete thought, so it is not an independent clause and does not get its own symbol. This statement should be represented by a single letter, such as s : The stripes on the U.S. flag are red and white.

EXAMPLE 2.9

Translating Compound Statements into Symbolic Form

Let p represent the statement, "It is a warm sunny day," and let q represent the statement, "the family will go to the beach." Write the symbolic form of each of the following compound statements.

1. If it is a warm sunny day, then the family will go to the beach.
2. The family will go to the beach, and it is a warm sunny day.
3. The family will not go to the beach if and only if it is not a warm sunny day.
4. The family not go to the beach, or it is a warm sunny day.

Solution

1. Replace "it is a warm sunny day" with p . Replace "the family will go to the beach." with q . Next, because the connective is *if ..., then* place the conditional symbol, $\rightarrow$, between p and q . The compound statement is written symbolically as: $p \rightarrow q$.
2. Replace "The family will go to the beach" with q . Replace "it is a warm sunny day." with p . Next, because the connective is *and*, place the $\wedge$ symbol between q and p . The compound statement is written symbolically as: $q \wedge p$.
3. Replace "The family will not go to the beach." with $\sim q$. Replace "it is not a warm sunny day" with $\sim p$. Next, because the connective is *or, if and only if*, place the biconditional symbol, $\leftrightarrow$ between $\sim q$ and $\sim p$. The compound statement is written symbolically as: $\sim q \leftrightarrow \sim p$.
4. Replace "The family will not go to the beach" with $\sim q$. Replace "it is a warm sunny day" with p . Next, because the connective is *or*, place the $\vee$ symbol between $\sim q$ and p . The compound statement is written symbolically as: $\sim q \vee p$.

> YOUR TURN 2.9

Let p represent the statement, "Last night it snowed," and let q represent the statement, "Today we will go skiing." Write the symbolic form of each of the following compound statements:

1. Today we will go skiing, but last night it did not snow.

2. Today we will go skiing if and only if it snowed last night.
3. Last night is snowed or today we will not go skiing.
4. If it snowed last night, then today we will go skiing.

Translating Compound Statements in Symbolic Form with Parentheses into Words

When parentheses are written in a logical argument, they group a compound statement together just like when calculating numerical or algebraic expressions. Any statement in parentheses should be treated as a single component of the expression. If multiple parentheses are present, work with the inner most parentheses first.

Consider your friend's struggles to get their license to drive. Let p represent the statement, "My friend passed the written test," let q represent the statement, "My friend passed the road test," and let r represent the statement, "My friend received a driver's license." The statement $(p \wedge q) \rightarrow r$ can be translated into words as follows: the statement $p \wedge q$ is grouped together to form the hypothesis of the conditional statement and r is the conclusion. The conditional statement has the form "if $p \wedge q$, then r ." Therefore, the written form of this statement is: "If my friend passed the written test and they passed the road test, then my friend received a driver's license."

Sometimes a compound statement within parentheses may need to be negated as a group. To accomplish this, add the phrase, "it is not the case that" before the translation of the phrase in parentheses. For example, using p , q , and r of your friend obtaining a license, let's translate the statement $\sim(p \wedge q) \rightarrow \sim r$ into words.

In this case, the hypothesis of the conditional statement is $\sim(p \wedge q)$ and the conclusion is $\sim r$. To negate the hypothesis, add the phrase "it is not the case" before translating what is in parentheses. The translation of the hypothesis is the sentence, "It is not the case that my friend passed the written test and they passed the road test," and the translation of the conclusion is, "My friend did not receive a driver's license." So, a translation of the complete conditional statement, $\sim(p \wedge q) \rightarrow \sim r$ is: "If it is not the case that my friend passed the written test and the road test, then my friend did not receive a driver's license."

 It is acceptable to interchange proper names with pronouns and remove repeated phrases to make the written statement more readable, as long the meaning of the logical statement is not changed.

VIDEO

[Logic Part 1B: Compound Statements, Connectives and Symbols \(https://openstax.org/r/Compound_Statements\)](https://openstax.org/r/Compound_Statements)

EXAMPLE 2.10

Translating Compound Statements in Symbolic Form with Parentheses into Words

Let p represent the statement, "My child finished their homework," let q represent the statement, "My child cleaned her room," let r represent the statement, "My child played video games," and let s represent the statement, "My child streamed a movie." Translate each of the following symbolic statements into words.

1. $\sim(p \wedge q)$
2. $(p \wedge q) \rightarrow (r \vee s)$
3. $\sim(r \vee s) \leftrightarrow \sim(p \wedge q)$

Solution

1. Replace $\sim$ with "It is not the case," and $\wedge$ with "and." One possible translation is: "It is not the case that my child finished their homework and cleaned their room."
2. The hypothesis of the conditional statement is, "My child finished their homework and cleaned their room." The conclusion of the conditional statement is, "My child played video games or streamed a movie." One possible translation of the entire statement is: "If my child finished their homework and cleaned their room, then they played video games or streamed a movie."
3. The hypothesis of the biconditional statement is $\sim(r \vee s)$ and is written in words as: "It is not the case that my child played video games or streamed a movie." The conclusion of the biconditional statement is $\sim(p \wedge q)$, which translates to: "It is not the case that my child finished their homework and cleaned their room." Because the biconditional, $\leftrightarrow$ translates to *if and only if*, one possible translation of the statement is: "It is not the case that my child played video games or streamed a movie if and only if it is not the case that my child finished their homework and cleaned their room."

> YOUR TURN 2.10

Let p represent the statement, “My roommates ordered pizza,” let q represent the statement, “I ordered wings,” and let r be the statement, “Our friends came over to watch the game.” Translate the following statements into words.

1. $\sim r \rightarrow (p \vee q)$
2. $(p \wedge q) \rightarrow r$
3. $\sim(p \vee r)$

The Dominance of Connectives

The order of operations for working with algebraic and arithmetic expressions provides a set of rules that allow consistent results. For example, if you were presented with the problem $1 + 3 \times 2$, and you were not familiar with the order of operation, you might assume that you calculate the problem from left to right. If you did so, you would add 1 and 3 to get 4, and then multiply this answer by 2 to get 8, resulting in an incorrect answer. Try inputting this expression into a scientific calculator. If you do, the calculator should return a value of 7, not 8.

[Scientific Calculator \(https://openstax.org/r/Scientific_Calculator\)](https://openstax.org/r/Scientific_Calculator)

The order of operations for algebraic and arithmetic operations states that all multiplication must be applied prior to any addition. Parentheses are used to indicate which operation—addition or multiplication—should be done first. Adding parentheses can change and/or clarify the order. The parentheses in the expression $1 + (3 \times 2)$ indicate that 3 should be multiplied by 2 to get 6, and then 1 should be added to 6 to get 7: $1 + (3 \times 2) = 7$.

As with algebraic expressions, there is a set of rules that must be applied to compound logical statements in order to evaluate them with consistent results. This set of rules is called the **dominance of connectives**. When evaluating compound logical statements, connectives are evaluated from least dominant to most dominant as follows:

1. Parentheses are the least dominant connective. So, any expression inside parentheses must be evaluated first. Add as many parentheses as needed to any statement to specify the order to evaluate each connective.
2. Next, we evaluate negations.
3. Then, we evaluate conjunctions and disjunctions from left to right, because they have equal dominance.
4. After evaluating all conjunctions and disjunctions, we evaluate conditionals.
5. Lastly, we evaluate the most dominant connective, the biconditional. If a statement includes multiple connectives of equal dominance, then we will evaluate them from left to right.

See [Figure 2.7](#) for a visual breakdown of the dominance of connectives.

Dominance	Connective	Symbol	Evaluate
Least Dominant	Parentheses	()	First
	Negation	$\sim$	↓
	Disjunction/Conjunction	$\vee, \wedge$	Left to right or add parentheses to specify order because or/and have equal dominance.
	Conditional	$\rightarrow$	↓
Most Dominant	Biconditional	$\leftrightarrow$	Last

Figure 2.7

Let's revisit your friend's struggles to get their driver's license. Let p represent the statement, “My friend passed the written test,” let q represent the statement, “My friend passed the road test,” and let r represent the statement, “My friend received a driver's license.” Let's use the dominance of connectives to determine how the compound statement $p \wedge \sim q \rightarrow r$ should be evaluated.

Step 1: There are no parentheses, which is least dominant of all connectives, so we can skip over that.

Step 2: Because negation is the next least dominant, we should evaluate $\sim q$ first. We could add parentheses to the statement to make it clearer which connecting needs to be evaluated first: $p \wedge \sim q \rightarrow r$ is equivalent to $p \wedge (\sim q) \rightarrow r$.

Step 3: Next, we evaluate the conjunction, $\wedge$. $p \wedge (\sim q) \rightarrow r$ is equivalent to $(p \wedge (\sim q)) \rightarrow r$.

Step 4: Finally, we evaluate the conditional, $\rightarrow$, as this is the most dominant connective in this compound statement.

?? WHO KNEW?

When using spreadsheet applications, like Microsoft Excel or Google Sheets, have you noticed that core functions, such as sum, average, or rate, have the same name and syntax for use? This is not a coincidence. Various standards organizations have developed requirements that software developers must implement to meet the conditions set by governments and large customers worldwide. The OpenDocument Format OASIS Standard enables transferring data between different office productivity applications and was approved by the International Standards Organization (ISO) and IEC on May 6, 2006.

Prior to adopting these standards, a government entity, business, or individual could lose access to their own data simply because it was saved in a format no longer supported by a proprietary software product—even data they were required by law to preserve, or data they simply wished to maintain access to.

Just as rules for applying the dominance of connectives help maintain understanding and consistency when writing and interpreting compound logical statements and arguments, the standards adopted for database software maintain global integrity and accessibility across platforms and user bases.

EXAMPLE 2.11

Applying the Dominance of Connectives

For each of the following compound logical statements, add parentheses to indicate the order to evaluate the statement. Recall that parentheses are evaluated innermost first.

1. $p \wedge \sim q \vee r$
2. $q \rightarrow \sim p \wedge r$
3. $\sim(p \vee q) \leftrightarrow \sim p \wedge \sim q$

✓ Solution

1. Because negation is the least dominant connective, we evaluate it first: $p \wedge (\sim q) \vee r$. Because conjunction and disjunction have the same dominance, we evaluate them left to right. So, we evaluate the conjunction next, as indicated by the additional set of parentheses: $(p \wedge (\sim q)) \vee r$. The only remaining connective is the disjunction, so it is evaluated last, as indicated by the third set of parentheses. The complete solution is: $((p \wedge (\sim q)) \vee r)$.
2. Negation has the lowest dominance, so it is evaluated first: $q \rightarrow (\sim p) \wedge r$. The remaining connectives are the conditional and the conjunction. Because conjunction has a lower precedence than the conditional, it is evaluated next, as indicated by the second set of parentheses: $q \rightarrow ((\sim p) \wedge r)$. The last step is to evaluate the conditional, as indicated by the third set of parentheses: $(q \rightarrow ((\sim p) \wedge r))$.
3. This statement is known as De Morgan's Law for the negation of a disjunction. It is always true. [Section 2.6](#) of this chapter will explore De Morgan's Laws in more detail.
 - First, we evaluate the negations on the right side of the biconditional prior to the conjunction.
 - Then, we evaluate the disjunction on the left side of the biconditional, followed by the negation of the disjunction on the left side.
 - Lastly, after completely evaluating each side of the biconditional, we evaluate the biconditional. It does not matter which side you begin with.

The final solution is: $(\sim(p \vee q)) \leftrightarrow ((\sim p) \wedge (\sim q))$.

> YOUR TURN 2.11

For each of the following compound logical statements, add parentheses to indicate the order in which to evaluate the statement. Recall that parentheses are evaluated innermost first.

1. $p \vee q \wedge \sim r$
2. $\sim p \rightarrow q \vee r$

$$3. \sim p \vee \sim q \leftrightarrow \sim(p \wedge q)$$

WORK IT OUT

Logic Terms and Properties – Matching Game

Materials: For every group of four students, include 30 cards (game, trading, or playing cards), 30 individual clear plastic gaming card sleeves, and 30 card-size pieces of paper. Prepare a list of 60 questions and answers ahead of time related to definitions and problems in [Statements and Quantifiers](#) and [Compound Statements](#). Provide each group of four students with 20 questions and their associated answers. Instruct each group to select 15 of the 20 questions, and then, for each problem selected, create one card with the question and one card with the answer. Instruct the groups to then place each problem and answer in a separate card sleeve. Once they create 15 problem cards and 15 answer cards, have students shuffle the set of cards.

To play the game, groups should exchange their card decks to ensure no team begins playing with the deck that they created. Split each four-person group into teams of two students. After shuffling the cards, one team lays cards face down on their desk in a five-by-six grid. The other team will go first.

Each player will have a turn to try matching the question with the correct answer by flipping two cards to the face up position. If a team makes a match, they get to flip another two cards; if they do not get a match, they flip the cards face down and it is the other team's turn to flip over two cards. The game continues in this manner until teams match all question cards with their corresponding answer cards. The team with the most set of matching cards wins.

In the first module of this chapter, we evaluated the truth value of negations. In the following modules, we will discuss how to evaluate conjunctions, disjunctions, conditionals, and biconditionals, and then evaluate compound logical statements using truth tables and our knowledge of the dominance of connectives.

Check Your Understanding

9. A _____ is a logical statement formed by combining two or more statements with connecting words, such as and, or, but, not, and if ..., then.
10. A _____ is a word or symbol used to join two or more logical statements together to form a compound statement.
11. The most dominant connective is the _____.
12. _____ are used to specify which logical connective should be evaluated first when evaluating a compound statement.
13. Both _____ and _____ have equal dominance and are evaluated from left to right when no parentheses are present in a compound logical statement.



SECTION 2.2 EXERCISES

For the following exercises, translate each compound statement into symbolic form.

Given p : "Layla has two weeks for vacation," q : "Marcus is Layla's friend," r : "Layla will travel to Paris, France," and s : "Layla and Marcus will travel together to Niagara Falls, Ontario."

1. If Layla has two weeks for vacation, then she will travel to Paris, France.
2. Layla and Marcus will travel together to Niagara Falls, Ontario or Layla will travel to Paris, France.
3. If Marcus is not Layla's friend, then they will not travel to Niagara Falls, Ontario together."
4. Layla and Marcus will travel to Niagara Falls, Ontario together if and only if Layla and Marcus are friends.
5. If Layla does not have two weeks for vacation and Marcus is Layla's friend, then Marcus and Layla will travel together to Niagara Falls, Ontario.
6. If Layla has two weeks for vacation and Marcus is not her friend, then she will travel to Paris, France.

For the following exercises, translate each compound statement into symbolic form.

Given p : "Tom is a cat," q : "Jerry is a mouse," r : "Spike is a dog," s : "Tom chases Jerry," and t : "Spike catches Tom."

7. Jerry is a mouse and Tom is a cat.
8. If Tom chases Jerry, then Spike will catch Tom.
9. If Spike does not catch Tom, then Tom did not chase Jerry.

10. Tom is a cat and Spike is a dog, or Jerry is not a Mouse.
11. It is not the case that Tom is not a cat and Jerry is not a mouse.
12. Spike is not a dog and Jerry is a mouse if and only if Tom chases Jerry, but Spike does not catch Tom.

For the following exercises, translate the symbolic form of each compound statement into words.

Given p : "Tracy Chapman plays guitar," q : "Joan Jett plays guitar," r : "All rock bands include guitarists," and s : "Elton John plays the piano."

13. $p \vee r$
14. $\sim s \rightarrow \sim q$
15. $(p \wedge q) \leftrightarrow r$
16. $\sim r \wedge (q \vee s)$
17. $\sim(p \wedge \sim q)$
18. $(q \rightarrow \sim r) \leftrightarrow (\sim p \vee \sim r)$

For the following exercises, translate the symbolic form of each compound statement into words.

Given p : "The median is the middle number," q : "The mode is the most frequent number," r : "The mean is the average number," s : "The median, mean, and mode are equal," and t : "The data set is symmetric."

19. $t \rightarrow s$
20. $p \wedge (q \wedge r)$
21. $\sim t \rightarrow \sim s$
22. $(r \wedge p) \leftrightarrow q$
23. $(t \rightarrow \sim q) \vee (r \rightarrow s)$
24. $\sim(q \vee r) \rightarrow t$

For the following exercises, apply the proper dominance of connectives by adding parentheses to indicate the order in which the statement must be evaluated.

25. $p \rightarrow q \vee r$
26. $p \wedge q \leftrightarrow \sim r$
27. $p \vee r \vee \sim q$
28. $p \wedge \sim q \wedge r$
29. $p \wedge r \vee s \wedge t$
30. $q \rightarrow \sim r \leftrightarrow \sim p \vee \sim r$
31. $p \rightarrow r \vee s \leftrightarrow \sim t$
32. $\sim(t \wedge s) \vee (p \rightarrow q) \wedge \sim r$